

Epitalon (AEDG) for Sleep **Available for Research Use Only**

(Epitalon nightly 100–300 mcg, Intranasal or subcutaneous)

Epitalon (AEDG) is a synthetic tetrapeptide originally developed as a standardized, reproducible analogue of the Russian pineal extract Epithalamin. While Epitalon is often discussed in the context of multi-axis longevity protocols, this document focuses exclusively on community use of Epitalon as a sleep and circadian rhythm–reset peptide, a practice that has emerged independently of the original Russian gerontology literature.

💉💉 Probably safe (some human evidence)

★ Plausibly effective (in-vitro and animal signals are good, weak human data)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.



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The evidence for synthetic Epitalon long-term human safety is **limited to reassuring animal signals, limited peer-reviewed human safety data, and frequent practitioner use reports** supporting safety, but there are some mechanistic/epigenetic concerns. No modern Phase-1/2 safety trials exist. Epithalamin has been in longer use and has a better evidence scope for human safety.)

The evidence for synthetic Epitalon efficacy for sleep is weak, **based only on moderate mechanistic plausibility, some supportive animal circadian data, and widespread, uncontrolled, mixed community reports.** (Epithalamin is more studied with better evidence.)

Replacing Epithalamin with Epitalon

IBG developed the synthetic peptide Epitalon (AEDG), to replace the bovine polypeptide extract (Epithalamin) so effects could be standardized, manufactured reproducibly, and studied mechanistically. IBG claims Epitalon reproduces Epithalamin's effects on **melatonin secretion and circadian regulation**.

Biological Mechanism

Epitalon acts primarily through the **pineal–circadian axis**, enhancing melatonin rhythmicity, stabilizing clock-gene expression, and supporting telomere-associated cellular longevity mechanisms.

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Epitalon is described as enhancing **melatonin rhythmicity**, supporting **pineal function**, stabilizing **clock-gene expression**, and improving **sleep timing** indirectly.

Preclinical work suggests Epitalon may influence **CLOCK/BMAL1** regulatory loops, improve **chromatin stability** in circadian-regulated genes, reduce **oxidative stress** in pinealocytes, and support **DNA repair pathways** linked to circadian regulation. These upstream actions may produce more consistent sleep onset, reduced nighttime fragmentation, improved morning alertness, and better alignment of sleep–wake cycles. The logic is that Epitalon may **nudge pineal signaling**, thus helping restore more stable melatonin rhythms.

These mechanisms are hypothesized, not proven in modern human trials.

Dosage & Patterns

Epitalon has no modern, well-controlled human trials establishing doses.

Longevity dosing information is presented as reference, but is not the focus of this research survey: The doses below are published in the **2002 Biogerontology paper: Khavinson VKh, Malinin VV. Epithalamin and Epitalon: peptide geroprotectors. Biogerontology. 2002;3(3):171-178.** For pineal/circadian axis reset

- Epitalon 5 – 10 mg IM daily
- For short daily courses, most often 10-days, sometimes 3-days, sometimes 20-days
- Repeated once or twice annually
- IBG administration is IM, practitioner dosing is usually SC, leading to unpredictable differences due to bioavailability.

Dose Controversy (Epitalon)

Russian researchers historically used Epithalamin (a pineal gland *extract*) in many animal and clinical studies. Later research studies and community practice uses **synthetic Epitalon**. These are related but **not identical**.

There is **no validated, universal conversion ratio** between extract doses and synthetic, pure peptide doses because

- Unknown and variable composition of Epithalamin extracts.
- Epithalamin's biological effects may arise from several peptides acting together.
- Matrix effects in Epithalamin (carrier peptides, salts, protease inhibitors) can alter absorption, proteolysis, and tissue exposure compared with a purified synthetic peptide.

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Some discussions create the impression that Epithalamin is “weaker” per mg, **but mass-based comparisons are misleading**. Because Epithalamin **doses are expressed as crude mass**, not quantified peptide, **a larger extract dose may still contain less active peptide than a small synthetic peptide dose**.

Claims that the extract is “weaker” are **plausible but unproven**; responsible translation requires quantification of active components and exposure-based PK/PD mapping.

Dose for Sleep

Russian Epitalon research **never describes or even hints at the idea of continuous low-dose Epitalon for sleep**. The Russian researchers consistently frame Epitalon as a peptide that **produces long-lasting downstream effects after the course is finished, rather than something meant to be taken nightly or continuously**. Epitalon’s biological mechanisms, as discussed by researchers, make **continuous use biologically incoherent**.

Russian studies do report that **sleep improvements occur because Epitalon normalizes melatonin rhythms and restores pineal function** after the short-course, but not because the peptide is taken at bedtime as a sleep-aid. The improvements are reported to **emerge gradually and persist for weeks or months** without further use.

Western FRU **low dosing of Epitalon for sleep** is a grassroots practice **built from anecdote, experimentation, and community lore** rather than from any peer-reviewed research or mechanistic biology. **Practitioners talk about Epitalon as if it were a nightly sleep-aid**, something taken in **100 – 300 mcg doses**, often intranasal, before bed to **smooth circadian rhythm, deepen sleep, or reduce nighttime awakenings**. **100 mcg is the most common dose**, with **200 – 300 mcg discussed for more difficult sleep problems**.

- **100 mcg nightly one hour before sleep**
- **300 mcg for more difficult sleep issues**
- **Taken ~1 hour before bed**
- **Sometimes cycled, sometimes continuous**

Any use of Epitalon microdosing for sleep falls completely self-experimentation, with the only guidance for patterns and doses coming from other individual self-experimenters.

Drivers of Diminished Response and Mitigation Strategies

For low-dose Epitalon, for receptor desensitization the risk is low because Epitalon acts on pineal and circadian pathways and is not a classical receptor agonist. For **neutralizing antibodies** the risk is low because Epitalon is a small tetrapeptide with limited immunogenicity reported in human studies. For **downstream pathway adaptation** circadian and transcriptional

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networks can adapt to chronic modulation and reduce effect over time. For **system level physiological adaptation** homeostatic circadian adjustments can blunt sustained benefit. **Overall likelihood of waning is low to moderate.**

Some users say the **effect fades with nightly use and require breaks**, while **others continue indefinitely**.

Combined Use

Some community users combine Epitalon with:

- **DSIP**
- **Melatonin**
- **Sleep hygiene protocols**
- **Stress-reduction routines**

These are community practices without any clinical, animal, or *in-vitro* evidence.

Benefits

- **Smoother sleep onset**
- **Gentle improvements in sleep continuity**
- **Reduced nighttime awakenings**
- **Better circadian stability after travel or stress**

These reports are **anecdotal**, not clinical evidence. Practitioners frame it as a **gentle, ongoing “somnific” support** rather than a therapeutic cycle. The logic, not framed in mechanistic biology, is that small, steady doses **might nudge melatonin production, stabilize sleep architecture, or maintain pineal signaling without the intensity of the Russian ten-to-twenty-day high-dose courses.**

Based on mechanistic arguments, Khavinson and IBG discuss normalization of circadian and melatonin rhythms, which might improve sleep timing and hormonal regulation.

Indications

- Difficulty maintaining sleep
- Irregular circadian rhythms
- Jet lag
- Shift-work disruption
- Age-related sleep fragmentation
- “Resetting” sleep after chronic stress or travel

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Side Effects

- No effect
- Mild headache
- Occasional vivid dreams

Cancer Risks

Because Epitalon influences pathways relevant to cell proliferation, cancer-related questions are common. Epitalon can **activate telomerase and lengthen telomeres in normal human cells**. Since some cancers rely on telomerase for immortalization, anything that boosts telomerase is *theoretically* pro-tumor. But, **preclinical data actually shows** in multiple animal cancer models, Epitalon has tended to **delay tumor appearance, reduce tumor number/size, or inhibit metastasis**, rather than accelerate cancer.

References

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6. **Kossoy G, Anisimov VN, Ben-Hur H, et al.** Effect of the synthetic pineal peptide Epitalon on spontaneous carcinogenesis in female C3H/He mice. *In Vivo*. 2006;20(2):[pages].